

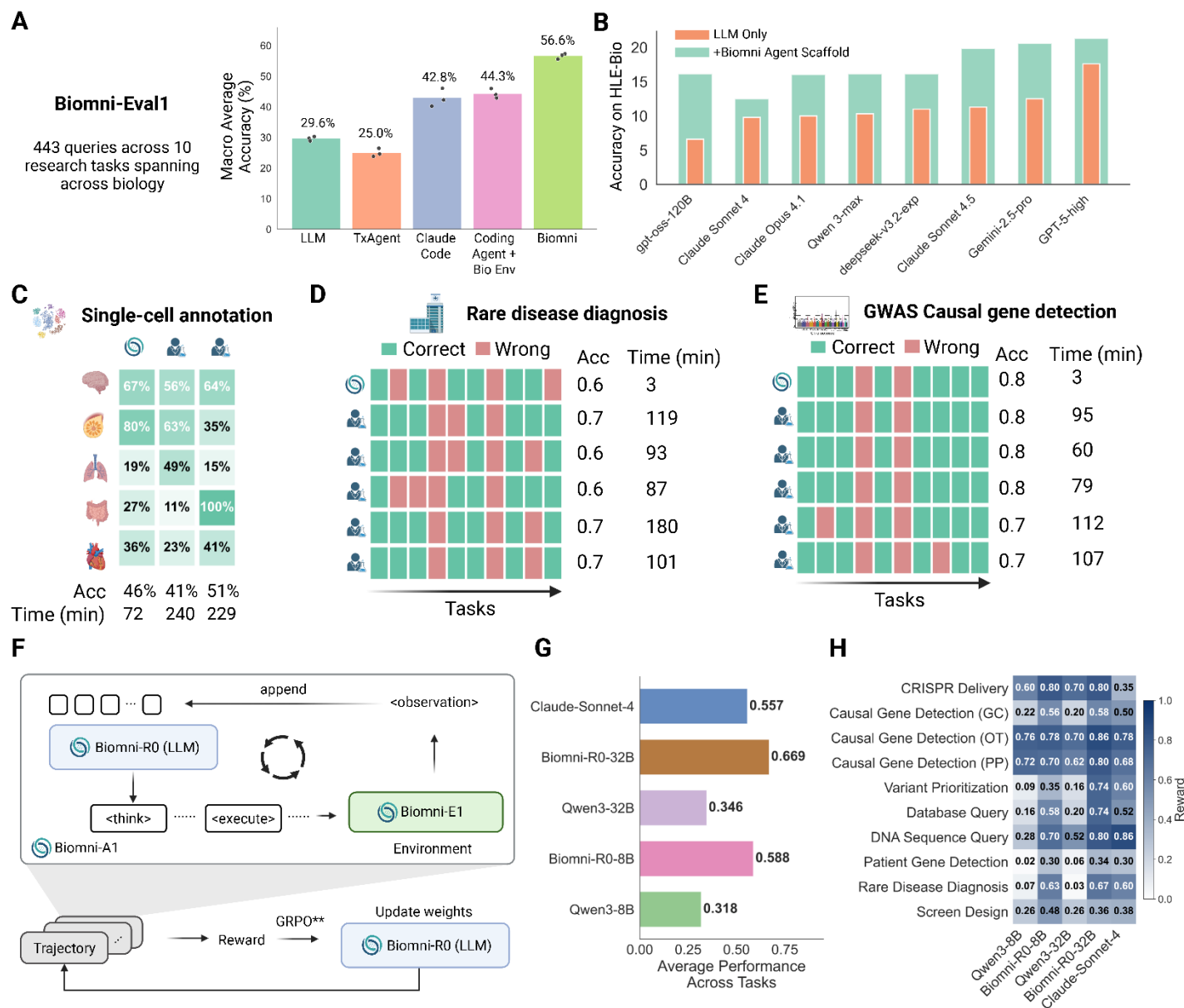


Fig. 2. Biomni performance across general benchmarks, expert-level tasks, and reinforcement learning. (A) Performance on Biomni-Eval1, comprising 443 queries across 10 biomedical research tasks, including CRISPR delivery, causal gene detection, variant prioritization, database and DNA sequence queries, patient gene detection, rare disease diagnosis, and screen design. The data points are 3 independent runs, and we report the average accuracy across 443 queries. (B) Accuracy on Humanity's Last Exam (HLE-Bio) across multiple frontier LLM backbones, comparing LLM-only baselines with the Biomni-A1 agent scaffold and Biomni-E1 agent environment. (C) Single-cell RNA-seq annotation accuracy and execution time compared with human experts, showing expert-comparable accuracy with reduced analysis time. (D) Rare disease diagnosis accuracy and execution time, with Biomni matching expert accuracy while substantially reducing time to completion. (E) GWAS causal gene detection accuracy and execution time, demonstrating expert-level performance with markedly faster execution. (F) Overview of the Biomni agent architecture and reinforcement learning training procedure. (G) Average task performance before and after reinforcement learning, showing substantial gains for Biomni-R0 models at 8B and 32B scales. Biomni-R0 was fine-tuned from Qwen3 open-weight model (37) and was first distilled from the frontier model Claude-Sonnet-4 (i.e., teacher model) and then performed reinforcement learning to further specialize to biomedical tasks (see materials and methods). It outperformed the teacher frontier model after tuning. (H) Task-level performance comparison across specialized biomedical tasks, illustrating consistent improvements from reinforcement learning across task categories.